Junghoon Chung

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Summary

Ph.D. student in Industrial & Systems Engineering with experience in UX research, human factors, and data-driven design. Designs mixed-methods studies that combine interviews, surveys, experiments, and behavioral or sensor data to inform trustworthy adaptive systems. Seeking UX Research internship opportunities.

Education

University of Washington, Seattle Sep 2025 – Jun 2029 (Expected)
Ph.D. in Industrial and Systems Engineering; Dean's Fellowship Recipient

Virginia Tech, Blacksburg Aug 2023 – May 2025
M.S. in Industrial and Systems Engineering (Thesis Track), GPA: 3.89/4.0

Yonsei University, Seoul Mar 2017 – Feb 2023
B.S. in Industrial Engineering, GPA: 3.52/4.0; Student Council President

Experience

Graduate Research Assistant, HAS Lab, University of Washington Sep 2025 – Present

- Design controlled experiments and user studies to evaluate attention, workload, and performance in complex task environments.
- Collect and analyze multimodal behavioral and physiological data to understand user states during sustained interaction and inform adaptive support.

Graduate Research Assistant, OEB Lab, Virginia Tech Aug 2023 – May 2025

- Led mixed-methods research on workplace technologies through interviews, surveys, behavioral observations, and sensor-based data collection.
- Synthesized qualitative feedback with longitudinal behavioral data to identify usability barriers and contextual drivers of comfort, productivity, and task engagement.

Selected Projects

Real-Time Vigilance Monitoring Interface *University of Washington*
Developing an EmotiBit-based interface that presents continuously updated vigilance estimates from heart-rate, electrodermal-activity, and photoplethysmography signals. Conducting user interviews to determine how, when, and how much monitoring information people want to see; planning lab and prospective classroom field evaluations of performance, usability, and perceptions of monitoring.

Driver Response to Vehicle-to-Everything Pedestrian Alerts *City of Bellevue / University of Washington*
Designing a field-study protocol with the City of Bellevue, T-Mobile, and Applied Information to examine how smartphone pedestrian alerts shape driver awareness and yielding decisions in naturalistic on-road conditions.

Phase-Based Modeling of Automated-Vehicle Takeover *University of Washington*
Developing a theory-grounded framework that examines pre-, during-, and post-takeover behavior to identify where drivers differ beyond a single takeover-success score.

NudgeWatch, Adaptive Sit-Stand Nudging System *University of Washington*
Designed and built a native Apple Watch intervention with real-time posture sensing and adaptive nudge logic. A five-participant mixed-methods pilot found that adaptation worked but was rarely perceived, informing more legible adaptive-system design.

Context-Aware Sit-Stand Intervention *Virginia Tech*
Conducted a three-week field study combining interviews, passive desk and computer logs, and individualized posture-transition models to identify when knowledge workers were receptive to personalized support.

Skills

Research: Mixed-methods user research, interviews, surveys, usability testing, experimental design
Analysis: Qualitative synthesis, behavioral data analysis, time-series analysis, evidence synthesis
Programming: Python, SQL (MySQL), R, Swift (watchOS)

Awards

Dean's Fellowship, College of Engineering — University of Washington, 2025–2029
Social-Innovation Capability Video Contest, 2nd Place — Yonsei University, 2022
Merit-based Scholarship (Academic Excellence) — Yonsei University, 2021